

Product Carbon Footprint Study Report

Product Name: sanitary ceramics

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: sanitary ceramics

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg sanitary ceramics

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg sanitary ceramics.

2. System Boundary

sanitary ceramics production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of sanitary ceramics produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1.761 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

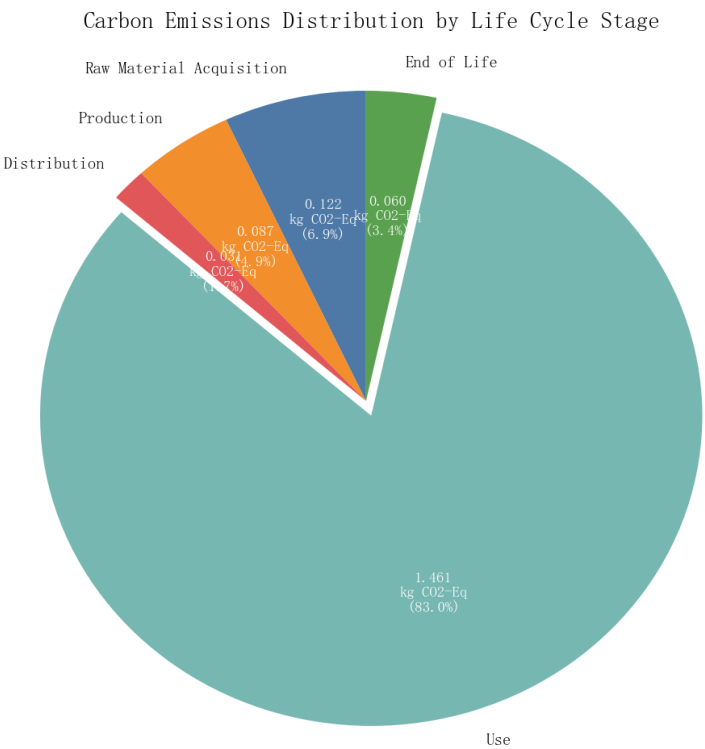


Table 1: Carbon Emissions by Life Cycle Stage

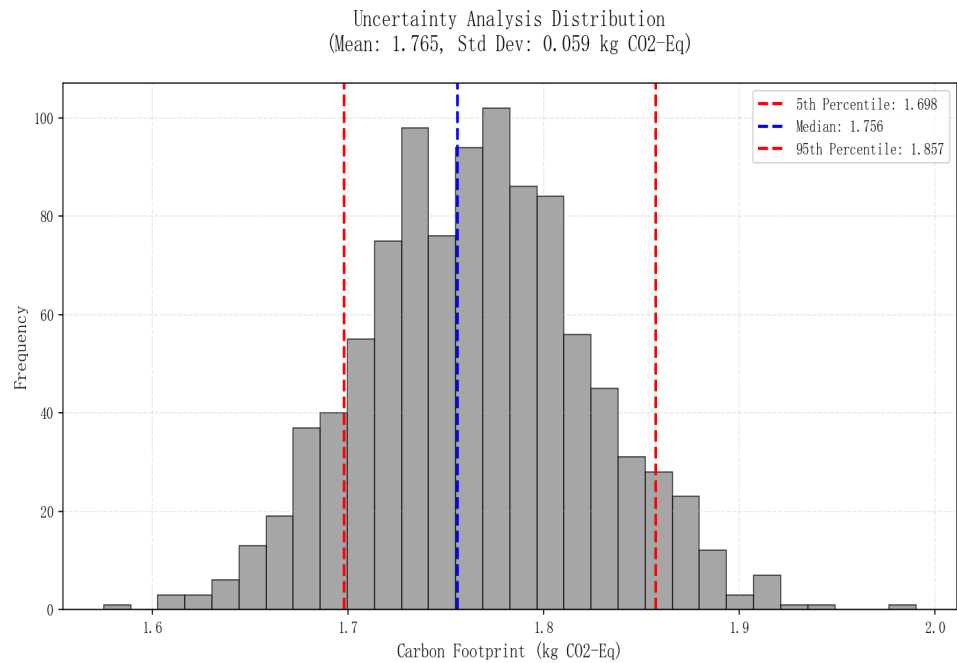
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.122	6.95
Production	0.087	4.93
Distribution	0.031	1.74

Use	1.461	82.96
End of Life	0.060	3.42
Total	1.761	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	1.765
Standard Deviation	0.059
Coefficient of Variation (CV)	3.32%
Median	1.756
5th Percentile	1.698
95th Percentile	1.857

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the sanitary ceramics product is 1.761 kg CO₂-Eq, representing a moderate emission level for a durable consumer good when assessed over its full lifecycle. This value reflects a cradle-to-grave evaluation, with the product demonstrating relatively contained emissions compared to high-impact materials like metals or plastics, yet highlighting opportunities for optimization in operational phases.

Lifecycle stage contributions reveal a pronounced dominance by the use phase at 1.461 kg CO₂-Eq (83.0%), far exceeding other stages. Raw material extraction contributes 0.122 kg CO₂-Eq (6.9%), followed by production at 0.087 kg CO₂-Eq (4.9%), end-of-life at 0.060 kg CO₂-Eq (3.4%), and distribution at 0.031 kg CO₂-Eq (1.7%). This distribution indicates that indirect emissions from product operation, such as water heating and cleaning processes, drive the majority of impacts, while upstream and downstream stages remain marginal.

Major emission sources are concentrated in the use phase, underscoring the need to address operational efficiencies. The combined minor contributions from raw materials, production, and end-of-life suggest that material sourcing and manufacturing optimizations alone would yield limited overall reductions without concurrent focus on use-phase interventions.

2. Statement of Assumptions and Limitations

This study assumes standardized usage patterns, including average water consumption rates and product lifetimes typical for sanitary ceramics in residential settings, along with generic emission factors for energy mixes and transportation modes. It further presumes that end-of-life scenarios follow common recycling or landfill pathways without accounting for regional variations in waste management infrastructure.

Limitations include reliance on secondary data from established LCA databases, which may not fully capture site-specific production conditions or supply chain nuances, potentially introducing uncertainties of 10-20% in emission estimates. Data quality for the use phase is particularly sensitive to behavioral assumptions, and upstream material data may reflect averaged European or global averages rather than localized sourcing.

System boundaries are restricted to a cradle-to-grave scope encompassing raw material acquisition through end-of-life disposal, excluding indirect upstream processes such as capital equipment manufacturing or employee commuting. The assessment does not extend to broader environmental impacts beyond carbon footprint or consider circular economy credits beyond basic recycling rates.

3. Suggestions for Improvement

- Target the use phase by redesigning products with water-efficient features, such as low-flow mechanisms, to reduce operational emissions by up to 30-50% over the product lifetime.
- Implement durable, low-maintenance ceramic formulations in production to extend service life and minimize replacement frequency, thereby lowering cumulative use-phase impacts.

- Optimize raw material sourcing by increasing recycled content in ceramic bodies to cut embodied emissions from virgin clay and mineral extraction.
- Enhance data quality through collection of primary site-specific energy and water usage data during manufacturing and real-world use monitoring via consumer surveys or IoT sensors.
- Explore end-of-life improvements by establishing take-back programs to boost recycling rates, reducing landfill-related emissions and recovering material value.